Calculus

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 18 cm. Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 11. Substituting $x = 2$ into $y = 2x + 7$ gives $y = 2(2) + 7 = 11$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 84°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 59^\circ) = 84^\circ$.

4 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

5 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A data set recorded for trigonometric identities consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

For the model $f(x) = 6x + 9$ in solving trigonometric equations 1, evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

7 **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A data set recorded for solving trigonometric equations 1 consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 2π radians about the x-axis.

Apply the volume of revolution formula: $V = \pi\int_0^4 y^2\,dx = \pi\int_0^4 (\sqrt{x})^2\,dx = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = \pi\left(\frac{16}{2} - 0\right) = 8\pi.$

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

[Calculus (HL) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \le \pi$. Hence find z^6 and determine all three solutions to $w^3 = z$.

1. Modulus: $r = |z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$. Argument: since z lies in Quadrant II, $\theta = \pi - \arctan(\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$. Polar form: $z = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$; Exponential form: $z = 2e^{i2\pi/3}$. 2. Compute z^6 via De Moivre's Theorem: $z^6 = 2^6 e^{i\cdot 6(2\pi/3)} = 64e^{i4\pi} = 64(\cos 4\pi + i\sin 4\pi) = 64(1 + 0i) = 64.$ 3. Solve $w^3 = z = 2e^{i2\pi/3}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2\cos x\,dx$.

1. First IBP: Let $u = x^2 \Rightarrow du = 2x\,dx$. Let $dv = \cos x\,dx \Rightarrow v = \sin x$. $\int_0^{\pi/2} x^2\cos x\,dx = [x^2\sin x]_0^{\pi/2} - 2\int_0^{\pi/2} x\sin x\,dx = \left(\left(\frac{\pi}{2}\right)^2(1) - 0\right) - 2\int_0^{\pi/2} x\sin x\,dx = \frac{\pi^2}{4} - 2J.$ 2. Second IBP on $J = \int_0^{\pi/2} x\sin x\,dx$: Let $u = x \Rightarrow du = dx$. Let $dv = \sin x\,dx \Rightarrow v = -\cos x$..

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{9^2 + 10^2 - 7^2}{2(9)(10)} = \frac{81 + 100 - 49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333.$ $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7 + 9 + 10}{2} = 13$..

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1-r} = \frac{12}{1-2/3} = \frac{12}{1/3} = 36$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Statistics & Probability

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD $6cmandtheradiusisUSD3$ cm, find the angle θ in radians.

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is USD 2 radians.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 AI SL — Specific Topics

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x-2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Number & Algebra

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$.
$y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Functions

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$. $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x - 4)^2 + (y + 3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4, -3)$ and radius $r = \sqrt{36} = 6$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Geometry & Trigonometry

(Paper 3 Extended Investigation) Consider the recursive Babylonian sequence defined by $x_0 = 3$ and $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$ for $n \geq 0$. (a) Use the AM-GM inequality to prove that $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Show that the sequence is strictly decreasing for $n \geq 1$, and deduce that $\lim_{n\to\infty} x_n = \sqrt{5}$. (c) Prove the quadratic error relationship $|x_{n+1} - \sqrt{5}| = \frac{(x_n - \sqrt{5})^2}{2x_n}$.

(a) Lower bound via AM-GM: For any positive real number $x_n > 0$, the Arithmetic Mean-Geometric Mean inequality states that $\frac{a+b}{2} \geq \sqrt{ab}$. With $a = x_n$ and $b = \frac{5}{x_n}$: $x_{n+1} = \frac{x_n + 5/x_n}{2} \geq \sqrt{x_n \cdot \frac{5}{x_n}} = \sqrt{5}$. Thus $x_n \geq \sqrt{5}$ for all $n \geq 1$. (b) Monotonicity & Convergence: Compute the difference: $x_{n+1} - x_n = \frac{1}{2}x_n + \frac{5}{2x_n} - x_n = \frac{5 - x_n^2}{2x_n}$. Since $x_n \geq \sqrt{5}$ for $n \geq 1$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Statistics & Probability

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

1. Non-parallel test: Direction vectors $\mathbf{d}_1 = \langle 1, -1, 2 \rangle$ and $\mathbf{d}_2 = \langle 2, 1, 1 \rangle$ are not scalar multiples ($1/2 eq -1/1$). 2. Normal direction perpendicular to both lines:

$\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, -1, 2; 2, 1, 1| = \mathbf{i}(-1 - 2) - \mathbf{j}(1 - 4) + \mathbf{k}(1 - (-2)) = -3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k} \propto (-1; 1; 1).$ Magnitude.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying exact values in right-angled triangles, $m\angle A = 35^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 100° ✓

b. 110°

c. 92°

d. 145°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 45^\circ) = 100^\circ$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in sine rule and the ambiguous case, evaluate $f(2)$.

a. 12 ✓

b. 15

c. 6

d. 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for sine rule and the ambiguous case consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the function $f(x) = x^3 - 6x^2 + 5$. The graph of $y = f(x)$ is shown below. Determine the coordinates of the local maximum of $f(x)$.

a. **(0, 5)** ✓

b. (4, -27)

c. (2, -11)

d. (1, 0)

To find the local maximum of the function $f(x) = x^3 - 6x^2 + 5$, we first need to find the critical points by taking the derivative and setting it to zero. 1. Find the first derivative:
$f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$ 2. Set the derivative to zero and solve for x: $3x^2 - 12x = 0$
$3x(x - 4) = 0$ This gives critical points at $x = 0$ and $x = 4$. 3. Use the second derivative test to determine if these points are local maxima or minima. Find the second derivative.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the function $f(x) = x^2 + k$ is shown below. The graph passes through the point $(3, 11)$. Determine the value of k.

a. **2** ✓

b. -2

c. 5

d. -5

The function is given by $f(x) = x^2 + k$. We are given that the graph of the function passes through the point $(3, 11)$. This means that when $x = 3$, $f(x) = 11$. Substitute these values into the function's equation: $11 = (3)^2 + k$ $11 = 9 + k$ To find k, subtract 9 from both sides of the equation: $k = 11 - 9$ $k = 2$ Therefore, the value of k is 2.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the function $f(x) = x^2 - 6x + 5$. Find the x-coordinate of the vertex of the parabola represented by $y = f(x)$.

a. 1

b. **3** ✓

c. -1

d. -3

The given function is a quadratic function in the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -6$, and $c = 5$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values of a and b, we get $x = -\frac{-6}{2(1)} = \frac{6}{2} = 3$. Therefore, the x-coordinate of the vertex is 3.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the function $f(x) = 2x^2 - 3x + 5$ is shown below. Determine the value of $f'(2)$.

a. **7** ✓

b. 5

c. 9

d. 2

The derivative of the function $f(x) = 2x^2 - 3x + 5$ is found using the power rule: $f'(x) = \frac{d}{dx}(2x^2) - \frac{d}{dx}(3x) + \frac{d}{dx}(5)$. This gives $f'(x) = 4x - 3$. To find $f'(2)$, substitute $x = 2$ into the derivative: $f'(2) = 4(2) - 3 = 8 - 3 = 5$.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. Find the y-intercept of the graph.

a. **3** ✓

b. -1

c. 0

d. -3

The y-intercept of a function is the value of the function when $x = 0$. For the function $f(x) = x^2 - 4x + 3$, we can find the y-intercept by substituting $x = 0$: \\ \(f(0) = (0)^2 - 4(0) + 3 \\ f(0) = 0 - 0 + 3 \\ f(0) = 3 \\ Thus, the y-intercept is 3.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^3 - 6x^2 + 5$. Find the values of x for which the tangent to the curve $y = f(x)$ is parallel to the x-axis.

a. $x = 0$ **or** $x = 4$ ✓

b. $x = 1$ or $x = 5$

c. $x = 0$ or $x = 6$

d. $x = -1$ or $x = -5$

The gradient of the tangent to the curve $y = f(x)$ is given by its derivative, $f'(x)$. If the tangent is parallel to the x-axis, its gradient is 0. First, find the derivative of $f(x) = x^3 - 6x^2 + 5$:
$f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$. Set the derivative equal to zero to find the x-values where the tangent is horizontal: $3x^2 - 12x = 0$. Factor out $3x$: $3x(x - 4) = 0$. This gives two possible values for x: $3x = 0 \Rightarrow x = 0$ $x - 4 = 0 \Rightarrow x = 4$. Therefore, the values of.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the equation of the tangent to $y = -2x^3 + 2x$ at $x = 1$, in the form $y = mx + c$.

a. $y = -4x + 6$

b. $y = -4x$

c. $y = 4x + 4$

d. $y = -4x + 4$ ✓

Working through this calculus step: $f'(x) = -6x^2 + 2 \Rightarrow m = f'(1) = -4$, $y_0 = f(1) = 0$. Tangent: $y + 0 = -4(x - 1) \Rightarrow y = -4x + 4$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the x-coordinates of the stationary points of $f(x)$, given $f'(x) = 2x^2 - 8x + 6$. Find the larger x-coordinate.

a. 2

b. 0

c. **3** ✓

d. 1

Working through this calculus step: Set $f'(x) = 0$: $2x^2 - 8x + 6 = 0 \Rightarrow x = 1$ or $x = 3$. Larger: 3.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the area enclosed by the curve $y = x^2$, the x-axis, and the line $x = 3$, for $0 \leq x \leq 3$.

a. **9** ✓

b. 5

c. 4

d. 6

Working through this calculus step: Area $= \int_0^3 x^2\,dx = [\frac{1}{3}x^3]_0^3 = 9$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the gradient of the tangent to $y = -4x^3 - 5x^2 - 5x + 8$ at $x = 0$.

a. 0

b. -15

c. 5

d. **-5** ✓

Working through this calculus step: $f'(x) = -12x^2 - 10x - 5 \Rightarrow f'(0) = -5$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Tangent to $y = x^3$ at $(2, 8)$: equation:

a. $y = 12x - 16$ ✓

b. $y = 12x + 8$

c. $y = 6x - 4$

d. $y = 8x - 8$

y'=3x²=12 at x=2; y-8=12(x-2) → y=12x-16.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the quadratic function $f(x) = x^2 - 8x + k$ has its vertex at the point $(h, 3)$. Determine the value of k.

a. 13

b. **19** ✓

c. 22

d. 25

The x-coordinate of the vertex of a quadratic function $f(x) = ax^2 + bx + c$ is given by $h = -\frac{b}{2a}$. For $f(x) = x^2 - 8x + k$, we have $a = 1$ and $b = -8$. Therefore, the x-coordinate of the vertex is $h = -\frac{-8}{2(1)} = \frac{8}{2} = 4$. We are given that the y-coordinate of the vertex is 3. This means $f(h) = f(4) = 3$. Substitute $x = 4$ into the function: $f(4) = (4)^2 - 8(4) + k$. So, $3 = 16 - 32 + k$. $3 = -16 + k$. Adding 16 to both sides, we get $k = 3 + 16 = 19$. Common algebraic trap: Students.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at (2, -4) and it passes through the point (4, 0). Determine the value of c.

a. **-4** ✓

b. 0

c. 2

d. 4

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is (2, -4), we have $f(x) = a(x - 2)^2 - 4$. \\ Since the graph passes through the point (4, 0), we can substitute these values into the equation to find 'a': \\ $0 = a(4 - 2)^2 - 4$ $0 = a(2)^2 - 4$ $0 = 4a - 4$ $4a = 4$ $a = 1$ Therefore, the function is $f(x) = 1(x - 2)^2 - 4$. \\ Expanding this equation to the form $f(x) = ax^2 + bx + c$: \\ $f(x) = (x^2 - 4x + 4) - 4$ $f(x) = x^2 - 4x$ Comparing this to.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown below. The graph has a minimum value of -5. Determine the value of k.

a. 4 ✓

b. 5

c. 14

d. 3

The minimum value of a quadratic function in the form $f(x) = ax^2 + bx + c$ occurs at the vertex. For $f(x) = x^2 - 6x + k$, the x-coordinate of the vertex is given by $-b/(2a) = -(-6)/(2*1) = 3$. \\ The minimum value of the function is $f(3)$. We are given that this minimum value is -5. \\ So, $f(3) = (3)^2 - 6(3) + k = -5$. \\ $9 - 18 + k = -5$ $-9 + k = -5$ $k = -5 + 9$ $k = 4$. \\ Common algebraic trap: Students might mistakenly set $f(0) = k$ as the minimum value, or confuse the x-coordinate of the vertex wit.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Functions

The graph of the quadratic function $f(x) = x^2 - 8x + k$ is tangent to the x-axis. The graph of the linear function $g(x) = -x + 1$ is also shown. Find the value of k.

a. 16 ✓

b. 8

c. 4

d. 2

For the quadratic function $f(x) = x^2 - 8x + k$ to be tangent to the x-axis, its discriminant must be zero. The discriminant of a quadratic $ax^2 + bx + c$ is $b^2 - 4ac$. In this case, $a = 1$, $b = -8$, and $c = k$. So, $(-8)^2 - 4(1)(k) = 0$, which means $64 - 4k = 0$. Solving for k, we get $4k = 64$, so $k = 16$. The value of k is 16. The graph of $g(x)$ is not needed to solve for k; it is provided to ensure the problem adheres to the problem archetype of using graphical information and potentially having dist.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Geometry & Trigonometry

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the sum of the x-coordinates of A and B.

a. -4

b. 3

c. **4** ✓

d. 1

For the graph of a quadratic function $f(x) = ax^2 + bx + c$, the sum of the roots (x-intercepts) is given by $-\frac{b}{a}$. In this case, $f(x) = x^2 - 4x + 3$, so $a = 1$ and $b = -4$. Therefore, the sum of the x-coordinates of the points where the graph intersects the x-axis is $-(\frac{-4}{1}) = 4$. Alternatively, we can find the roots by setting $f(x) = 0$: $x^2 - 4x + 3 = 0$. Factoring the quadratic gives $(x - 1)(x - 3) = 0$. The roots are $x = 1$ and $x = 3$. The sum of the roots is $1 + 3 = 4$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Statistics & Probability

Consider the function $f(x) = x^2 - 4x + 3$. Find the sum of the x-intercepts of the graph of $y = f(x)$.

a. **4** ✓

b. 3

c. -4

d. 1

The x-intercepts of the graph of $y = f(x)$ occur when $f(x) = 0$. So, we need to solve the quadratic equation $x^2 - 4x + 3 = 0$. This equation can be factored as $(x - 1)(x - 3) = 0$, which gives the x-intercepts as $x = 1$ and $x = 3$. The sum of the x-intercepts is $1 + 3 = 4$. Alternatively, for a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this case, $a = 1$, $b = -4$, and $c = 3$. Thus, the sum of the roots is $-\frac{-4}{1} = 4$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 AI SL — Specific Topics

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?

a. 1

b. 2

c. **3** ✓

d. 4

The points where the graph intersects the x-axis are the roots of the quadratic equation $f(x) = 0$. We need to solve $x^2 - 4x + 3 = 0$. Factoring the quadratic, we get $(x - 1)(x - 3) = 0$. Thus, the roots are $x = 1$ and $x = 3$. Let point A have coordinates $(1, 0)$ and point B have coordinates $(3, 0)$. The distance between A and B is the absolute difference of their x-coordinates: $|3 - 1| = 2$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Number & Algebra

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(5)$.

a. **8** ✓

b. 1

c. 0

d. 18

To find the value of $f(5)$, we substitute $x = 5$ into the function $f(x) = x^2 - 4x + 3$. $f(5) = (5)^2 - 4(5) + 3$ $f(5) = 25 - 20 + 3$ $f(5) = 5 + 3$ $f(5) = 8$ Therefore, the value of $f(5)$ is 8. Common errors might arise from miscalculating the square, the subtraction, or the addition.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Functions

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.

a. **2** ✓

b. -2

c. 1

d. 3

The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -b/(2a)$. In this case, $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -(-4)/(2 * 1) = 4/2 = 2$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Geometry & Trigonometry

A particle moves in the xy-plane. Its velocity vector $\mathbf{v}(t)$ at time t is given by $\mathbf{v}(t) = \langle 3t^2 - 6t, e^t - 1 \rangle$, where $t \geq 0$. The initial position of the particle at $t = 0$ is $\mathbf{r}(0) = \langle 5, -2 \rangle$. Find the position vector $\mathbf{r}(t)$ of the particle at time t. Then, determine the value of t for which the particle's speed is minimized and state the minimum speed.

a. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{2}$ at $t = 1$

b. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{5}$ at $t = 0$

c. $\mathbf{r}(t) = \langle 3t^2 - 6t + 5, e^t - t - 2 \rangle$, minimum speed 0 at $t = 0$

d. $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$**, minimum speed** $\sqrt{3}$ **at** $t = 0$ ✓

To find the position vector $\mathbf{r}(t)$, we integrate the velocity vector $\mathbf{v}(t)$ with respect to t. $\mathbf{r}(t) = \int \mathbf{v}(t)dt = \int \langle 3t^2 - 6t, e^t - 1 \rangle dt$ $\mathbf{r}(t) = \left\langle \int (3t^2 - 6t)dt, \int (e^t - 1)dt \right\rangle$ $\mathbf{r}(t) = \langle t^3 - 3t^2 + C_1, e^t - t + C_2 \rangle$ Using the initial condition $\mathbf{r}(0) = \langle 5, -2 \rangle$: For the x-component: $0^3 - 3(0)^2 + C_1 = 5 \Rightarrow C_1 = 5$ For the y-com.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Statistics & Probability

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

a. $y = 2x - 5$ ✓

b. $y = -\frac{1}{2}x + 5$

c. $y = 2x - 8$

d. $y = -2x + 11$

In a Voronoi diagram, the boundary edge between two sites is along the perpendicular bisector of the line segment connecting them. 1. Find the midpoint M of the segment AB: $M = \left(\frac{2+6}{2}, \frac{4+2}{2}\right) = (4, 3)$ 2. Find the gradient of segment AB: $m_{AB} = \frac{2-4}{6-2} = \frac{-2}{4} = -\frac{1}{2}$ 3. The gradient of the perpendicular bisector is the negative reciprocal of m_{AB}: $m_{\perp} = -\frac{1}{-\frac{1}{2}} = 2$ 4. Use the point-slope form with midpoint $(4, 3)$ and.

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